

Submission metadata

Methods in Ecology and Evolution

Manuscript: Auditing spatial information leakage in ecological prediction: quantifying the optimism in map accuracy with the **spLeakage** R package

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Impact statement

Short version (~30 words, for a tight character limit):

Ecological maps routinely report accuracy that is too optimistic because nearby survey points leak information between training and testing. We provide a free tool to detect, quantify and correct this overconfidence in any analysis or published map.

Extended version (~80 words, if a longer field is available):

Maps of where species occur, how much carbon a forest stores, or where disease risk is highest are only as trustworthy as their reported accuracy—yet that accuracy is routinely overstated, because spatially close survey points leak information between training and test data. This paper introduces an open-source R package that takes any existing analysis, or even a published map, and reports how inflated its accuracy is, where the leakage lies, and how to correct it—showing across 16 real datasets that reported accuracy is optimistic every time.

Suggested reviewers

The following researchers have directly relevant expertise in spatial cross-validation, map-accuracy assessment and species distribution modelling. They are chosen to span both perspectives the manuscript reconciles—the nearest-neighbour-matching / spatial-CV view and the design-based view—so the reconciliation is judged fairly. (*Please verify current affiliation and email at submission; none is a co-author and there are no collaborative conflicts of interest.*)

1. **Prof. Hanna Meyer** — University of Münster, Germany.
Lead developer of **CAST**, the area-of-applicability method and **NNDM/kNNDM**; the closest expert on the methods this paper builds on and audits.
2. **Prof. Alexandre M. J.-C. Wadoux** — INRAE / LISAH, Montpellier, France.
Leading proponent of design-based map-accuracy assessment; ideal to test that the paper's reconciliation treats the design-based ("spatial CV is pessimistic") case fairly.
3. **Prof. Alexander Brenning** — Friedrich Schiller University Jena, Germany.
Author of **sperrorest** and long-standing work on spatial cross-validation and spatial prediction error; strong on the statistical machinery.
4. **Dr. Roozbeh Valavi** — (SDM / spatial modelling; **blockCV** author).
First author of **blockCV** and of the presence/absence dataset used in the lead case study; expert on spatial blocking for species distribution models.
5. **Prof. Damaris Zurell** — University of Potsdam, Germany.
Leads work on species-distribution-model evaluation and reporting standards (ODMAP); well placed to judge the paper's relevance to ecological reporting practice.
6. **Prof. Gurutzeta Guillera-Arroita** — (quantitative ecology / SDM methods).
Expert on species distribution model evaluation and detectability; a methods-focused ecological

reviewer independent of the spatial-CV development groups.

Note on balance. If the editor prefers to limit reviewers who are themselves authors of the specific methods extended here (Meyer for NNDM/AOA; Valavi for **blockCV**), Brenning, Wadoux, Zurell and Guillerá-Arroita provide a fully independent yet equally expert panel.

Opposed reviewers. None.